

git log command

- > 1. The `git log` command in Git is used to display the commit history of a repository.
- 2. It shows a list of all the commits that have been made the repository, along with information like the commit hash, author, date and commit message.
- 3. Ones "`git log`" command executed it will display the commit history in **reverse Chronological order**(most recent commit first)

-> **Syntax:** `git log <options>`

Here options are optional.

-> If we execute "`git log`" command without any option then in the output we are getting below information like.

- 1) commit hash: -It 40 digit unique alphanumeric hash code.
- 2) Author: Who has committed the changes
- 3) Date: Date and time of commit
- 4) Commite Message

-> **Note:** If we want view all commits one by one then press **Enter key**.

-> **Note:** If we want to exit from "`git log`" command then press "**q**" key.

Options of git log command

1) **--oneline** : Display each commit in single line

Syntax:

`git log --oneline`

2) **--graph** : show the graphical representation of commit history

Syntax:

`git log --graph`

3) `--stat` : Display summary of changes introduced in each commit.

Syntax:

```
git log --stat
```

4) `--since` and `--until` : Display commit history for specified time duration

Syntax:

```
git log --since="2 week ago"
```

```
git log --since="2 day ago"
```

```
git log --until="2024-08-23"
```

5) `-p` or `--patch`: show actual changes(diff) introduced in each commit.

Syntax:

```
git log -p
```

or

```
git log --patch
```

6) `-n` : To limit the number of commit displayed.

Syntax:

```
git log -n <line_count>
```

Example:

```
git log -n 5
```

Display first 5 lines of the commit history

7) We can combine options in one command

```
git log --oneline -n 4
```

git revert command

- > **Usage:** The git revert command in Git is used to create a new commit that undoes the changes made by a previous commit.
- > "**git revert**" is a safe way to undo changes because it preserves the history and creates a new commit that effectively undo the changes introduced by the original commit.
- > **Syntax:**
git revert <commit-hash>

Example: git revert ad02aab
- 1. Once above command is executed, **first it will open text editor for you to enter commit message.**
- 2. create a new commit id and revert(undo) the changes of specified commit.
- 3. Lastly, commit(save) the changes.

options of "git revert" command

1. **--no-commit:** stage the changes for reverting but doesnot perform the autocommit.

This option allows you to review the changes or modify the changes before committing

Syntax: git revert --no-commit <commit-hash>

Example: git revert --no-commit ad02aab

Note: instead of '**--no-commit**' option we can use **-n** option

2. **--no-edit**: If we use this option then text editor will not open and use default commit message.

Syntax: `git revert --no-edit <commit-hash> ad02aab`

Example: `git revert --no-edit ad02aab`

3. We can specify the range of commit ids

Syntax: `git revert <start-commit> .. <end-commit>`

Example: `git revert ad02aab .. ad02ccd`

here two dots are required
while specifying
range

